

Tae Yong Song

Ph.D. Student in Building Science

Building Simulation Lab (BS Lab)

Department of Architecture and Architectural Engineering, Seoul National University

Website | Google Scholar | GitHub | LinkedIn

EDUCATION

- 2026.03–present** **Ph.D. in Architecture and Architectural Engineering**, Seoul National University
Advisor: Prof. Cheol-Soo Park
- 2022.03–2024.02** **M.S. in Architecture and Architectural Engineering**, Seoul National University
Advisor: Prof. Cheol-Soo Park
Thesis: “Exploring Potential of Transfer Learning in Overcoming Data Scarcity for Building Energy Model”
- 2015.03–2022.02** **B.S. in Architecture and Architectural Engineering**, Seoul National University

PUBLICATIONS

International Journal Articles

- 2025** Physics-embedded hybrid modelling approach for room temperature prediction using Siamese neural network and RC model
Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, Junu Lee, and Cheol-Soo Park. *Journal of Building Performance Simulation*, 19(2), 277–288.

International Conference Papers

- 2026** Curated Intelligence for Visual Inference and Deterministic Logic toward Automatic BEM
Tae Yong Song, Jin-Hong Kim, Seung-Ju Lee, Soo-Kyung Kim, Hyun-Chul Park, Kyung-Jae Kim, and Cheol-Soo Park. *Proceedings of the 7th Building Simulation Applications Conference*, pp. ??–??, June 24–26, Bozen-Bolzano, Italy.
- 2024** Physics-Informed Hybrid Modeling Approach for Room Temperature Prediction Using an RC Model and Siamese Neural Network
Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, and Cheol-Soo Park. *SimBuild 2024*.
- 2023** Causality-based model for decision-making of building energy retrofit
Chul-Hong Park, Young-Sub Kim, **Tae Yong Song**, Seon-Jung Ra, and Cheol-Soo Park. *Building Simulation 2023*, 1433–1438.
- 2023** Limitations of Conventional Artificial Neural Network-Based Surrogate Models for Building Energy Retrofit
Chul-Hong Park, Young-Sub Kim, Seon-Jung Ra, **Tae Yong Song**, and Cheol-Soo Park. *ASHRAE Annual Conference*, 683–690.

Domestic Conference Papers

- 2026** Development of Customized AI Technologies for BIM-to-BEM Automation: Integration of LLMs, Computer Vision, Delaunay Geometry Algorithms, and Minimum Cycle Algorithms
Tae Yong Song, Jin-Hong Kim, Soo-Kyung Kim, Hyun-Chul Park, Kyung-Jae Kim, and Cheol-Soo Park. *Proceedings of the 2026 Spring Annual Conference of the Architectural Institute of Korea*, 46(1), 829–830. **Outstanding Presentation Award.**
- 2022** Dynamic Sensitivity Analysis Using NODE-GAM
Tae Yong Song, Chul-Hong Park, Seon-Jung Ra, Young-Sub Kim, and Cheol-Soo Park. *Proceedings of the 2022 Annual Conference of the Korean Institute of Architectural Sustainable Environment and Building Systems*, 181–182.

RESEARCH PROJECTS

2026.03–present	Virtual Building Modeling and Simulation Framework, and a Causality-Based AR Tool for Net-Zero Building Design Evaluation <i>Korea Institute of Civil Engineering and Building Technology (KICT)</i>
2025.12–2026.01	IDF Model Development for Digital-Twin Energy Simulation <i>Eterion (Samsung Electronics)</i>
2025.10–2026.04	LLM Technology for Digital-Twin-Based Building Energy Models <i>Samsung Electronics</i>
2023.05–2023.11	Data-Driven HVAC Load Prediction and Optimal Air-Handling-Unit Control <i>LG Electronics</i>
2022.12–2023.07	Greenhouse-Gas Reduction Factors and Emissions Assessment for Building Green Remodeling <i>Ministry of Land, Infrastructure and Transport (MOLIT)</i>
2022.04–2024.12	Automated Digital Diagnosis and Design Technologies for Green Remodeling <i>Korea Agency for Infrastructure Technology Advancement (KAITA)</i>

TEACHING EXPERIENCE

2026–present	Teaching Assistant , Seoul National University Department of Architecture and Architectural Engineering, Seoul National University.
2023.03–2024.02	Teaching Assistant , Seoul National University Department of Architecture and Architectural Engineering, Seoul National University.

HONORS

2026	Outstanding Presentation Award Proceedings of the 2026 Spring Annual Conference of the Architectural Institute of Korea, Vol. 46, No. 1
------	---